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# Modeling the Light-Curve of Transiting Exoplanet WASP-135b

## AST325: Observational Astronomy

E. Shen

Department of Chemical and Physical Sciences, University of Toronto Mississauga

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**ABSTRACT**

The transit method is a primary tool for exoplanet detection, but extracting precise characteristics from ground based observations is highly dependent on photometric data quality. This study aims to develop a comprehensive Python based data reduction and modeling pipeline to extract the light-curve of the Hot Jupiter WASP-135b from raw telescope images and derive its physical and orbital parameters. We applied standard image calibration techniques, including master dark and flat fielding, followed by fixed aperture photometry to extract relative stellar flux. A Markov Chain Monte Carlo (MCMC) algorithm was subsequently utilized to fit a transit model to the resulting light-curve and estimate posterior probability distributions for key planetary parameters. The pipeline successfully recovered a planetary radius ratio of $R_p/R_s$=0.12, aligning with literature values. However, significant discrepancies were observed in the transit duration, limb darkening coefficients and transit center when compared to American Association of Variable Star Observers (AAVSO) reference data. These deviations were primarily attributed to high photometric scatter, incomplete transit coverage (missing ingress), and the use of uncorrected temporal frameworks. Subsequent application of the pipeline to the reference AAVSO dataset yielded parameter distributions highly consistent with published literature, confirming the computational integrity of the modeling code. While the fundamental transit depth is recoverable from noisy data, the constraint of orbital geometry and limb darkening requires high signal to noise photometry with a complete transit cycle.

## 1. Introduction

Although exoplanets sound extravagant, they are simply planets located outside of our solar system. To date, more than 6,000 exoplanets have been discovered, but fewer than 100 have been directly imaged. Because most exoplanets are too faint to be seen against the glare of their host stars, indirect detection techniques, such as the transit method, are predominantly used (Christiansen et al. 2025). When an exoplanet's orbit is nearly edge-on relative to our line of sight, the planet periodically passes in front of its host star, known as a transit. During this transit, the planet blocks a portion of the star's light, creating a measurable dip in the observed flux. This flux can then be modeled over time to produce a light-curve which provides critical information about the star-planet system.

WASP-135b is a gas giant first discovered by the Wide Angle Search for Planets (WASP) survey in 2016. Classified as a "Hot Jupiter", it belongs to a category with masses similar to Jupiter but with orbital periods under 10 days. It has a planetary mass, radius and orbital period of $1.90 \pm 0.08 M_{\rm Jup}$, $1.30 \pm 0.09 R_{\rm Jup}$ and 1.401 days, respectively. The planet orbits WASP-135, a G5V main-sequence star with properties similar to our Sun. It is located 980 light-years away in the Hercules constellation with a stellar mass and radius of $0.98 \pm 0.06 M_\odot$ and $0.96 \pm 0.05 R_\odot$, respectively (Spake et al. 2016).

While the detection of exoplanets has become a cornerstone of modern astrophysics, the true challenge lies in the high-precision characterization of the data. WASP-135b provides an ideal testbed for such analysis. Its large transit depth and short period allow for the development of data pipelines capable of extracting precise physical parameters for entry level astronomy students.

This paper aims to present a reproducible method in analyzing stellar datasets using Python code that is simple yet robust enough to produce values similar to literature. A series of Jupyter notebooks were produced to guide the audience through the steps to fully model a light-curve starting from raw telescope images.

## 2. Data & Methods

### 2.1. Data Reduction

The WASP-135b data used in this paper was supplied by Professor Lea Hirsch, whose contributions are greatly appreciated.

When data is collected through a telescope, the instrument induces an artificial digital "noise" caused by the camera heat and hardware imperfections. By subtracting the telescope sensor's internal noise (dark) and correcting for uneven lighting (flats), we isolate the actual light coming from the host star.

To isolate the celestial signal, we apply standard image calibration techniques. First, we created a master dark frame by median combining a series of dark exposures (images taken with the shutter closed at the same exposure time and temperature as the science frames) to remove thermal noise and map hot pixels. We then generated a master flat field by median combining flat exposures to correct for uneven pixel illumination. We calculated the calibrated image, $I_{cal}$, pixel-by-pixel using the raw image ($I_{raw}$), the master dark ($D$), and the normalized master flat ($F$):

$$I_{cal} = \frac{I_{raw} - D}{F} \tag{1}$$

### 2.2. Image Alignment & Aperture Photometry

As science images are captured over long periods of time, Earth's rotation will cause the telescope images to suffer from

tracking drifts. Aligning all our science images to a common coordinate system allows exact extraction of the total light emitted by the target and reference stars in each reduced frame.

We utilize the `astroalign` package developed by Martin Beroiz (Beroiz et al. 2020), which finds similar asterisms between images and estimates an appropriate transformation between them. The Central Limit Theorem is then applied to mitigate random Gaussian noise and improve the Signal-to-Noise Ratio (S/N). By grouping $n$ sequential calibrated images together, we successfully reduce scatter by a factor of $\sqrt{n}$.

Once aligned, we extracted stellar flux using aperture photometry. We placed a fixed circular aperture, with a radius determined by the Full Width at Half Maximum of the Point Spread Function, over the stars. To ensure we only measure stellar photons, local background sky noise is estimated using an annulus surrounding the star and subtracted.

### 2.3. Normalization & Error Propagation

Even after calibration, the light-curve will show variations caused by Earth's shifting atmosphere and weather. Normalization removes these atmospheric effects by comparing the target star to a constant baseline of nearby reference stars.

Three reference stars of similar brightness to the target are selected. The light-curve of each reference star is scaled to match the out-of-transit flux of the target. These scaled reference light-curves are then averaged together to create an "ensemble" artificial comparison star that models the atmospheric interference for that night. The target star's flux is then divided by this ensemble to produce the normalized relative light-curve:

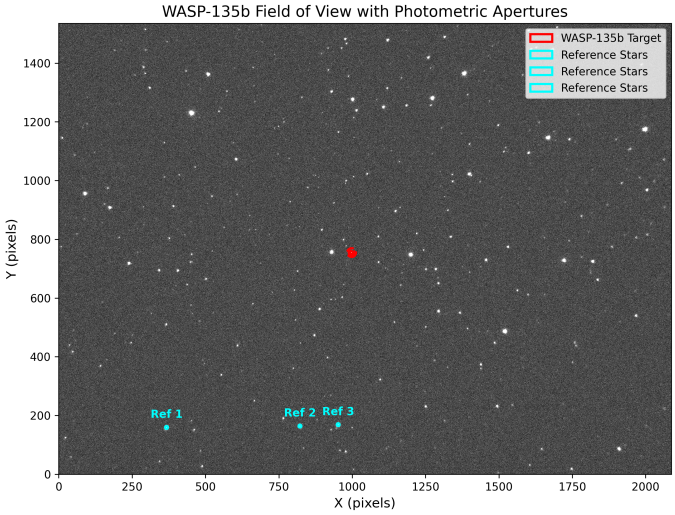



**Fig. 1:** Annotated star field of the aligned images. WASP-135b target star is highlighted in red, while the reference stars are marked in cyan

### 2.4. Transit Modeling

To extract physical parameters from the normalized data, we must fit a theoretical transit model to the light-curve and find the parameters that best match our observations.

To generate our model, we utilize the `batman` package developed by Laura Kreidberg (Kreidberg 2015). This model is formed by a set of parameters that determine the geometry of the transiting exoplanet as time passes. It utilizes the following parameters:

**Table 1:** Coordinates of target and reference stars with similar brightness

| Star | x Coordinate | y Coordinate |
|---|---|---|
| WASP-135 | 998.1 | 754.6 |
| Ref Star 1 | 367.5 | 159.1 |
| Ref Star 2 | 822.0 | 163.7 |
| Ref Star 3 | 952.3 | 168.5 |

**Notes.** "Similar" brightness was deemed to be fluxes within 50% to 200% of the target star

- $t_0$: Time of inferior conjunction (the center of the transit)
- $P$: Orbital period
- $R_p/R_s$: Ratio of the planet's radius to the star's radius (transit depth)
- $a/R_s$: Semi-major axis scaled by the stellar radius (transit duration)
- $i$: Orbital inclination in degrees (impact parameter and transit shape)
- $e$: Orbital eccentricity
- $\omega$: Longitude of periastron in degrees
- $u_1, u_2$: Limb darkening coefficients

### 2.5. Maximum Likelihood Estimation (MLE)

The likelihood is a measurement of how well our theoretical model fits the actual data. Log-likelihood values closer to zero represents a better fit between the model and the data. The best fit parameters are found by minimizing the negative log-likelihood using scipy.optimize (Virtanen et al. 2020).

Because our data points have associated measurement errors that are normally distributed, we use the Gaussian log-likelihood function. Instead of calculating the likelihood directly (possibility of extremely small numbers), the natural logarithm of the likelihood is calculated instead. The log-likelihood function is defined as:

$$\ln(\mathcal{L}) = -\frac{1}{2}\sum_{i=1}^{N}\left[\frac{(y_i - y_{exp,i})^2}{\sigma_i^2} + \ln(2\pi\sigma_i^2)\right] \quad (2)$$

where $y_i$ is the observed flux data, $y_{exp,i}$ is the generated model flux and $\sigma_i$ is the uncertainty of the observed flux, all at time $i$.

### 2.6. Bayesian Parameter Estimation via MCMC

While MLE provides a single "best fit," Bayesian analysis provides a full probability distribution for every parameter, allowing us to quantify uncertainties and incorporate prior physical knowledge. Bayesian inference relies on Bayes' Theorem to sample the posterior distribution:

$$P(\theta|D) \propto P(D|\theta)P(\theta) \quad (3)$$

where $P(\theta|D)$ is the posterior, $P(D|\theta)$ is the likelihood (from Section 2.5), and $P(\theta)$ represents our prior constraints.

We explore this parameter space using a Markov Chain Monte Carlo (MCMC) algorithm. A swarm of "walkers" traverses the likelihood space, mapping out the multi-dimensional parameter distributions. Convergence of the chains is verified using diagnostic trace plots. The final physical parameters and their $1\sigma$ uncertainties are extracted from the 16th, 50th (median), and 84th percentiles of the marginalized posterior distributions, which are visualized using a corner plot. We employ

the `emcee` Python package (Foreman-Mackey et al. 2013) for MCMC analysis and `corner.py` package for corner plotting (Foreman-Mackey 2016).

## 3. Results

A series of four Jupyter notebooks were developed with fellow AST325 classmates to develop this transit modeling pipeline. The WASP-135b data used was captured on July 14, 2024 from 4:45 - 8:28 and contained a total of 3357 transit frames and 20 dark frames. It is important to note that Jupyterhub has a memory limit, which if surpassed, will cause the kernel to become unresponsive. Thus, due to our data's high image count, a locally run Python interpreter, such as Pycharm, was utilized over the cloud based Jupyterhub to avoid this memory issue. To visualize the transit event, we plotted the relative flux of WASP-135b (Figure 2).

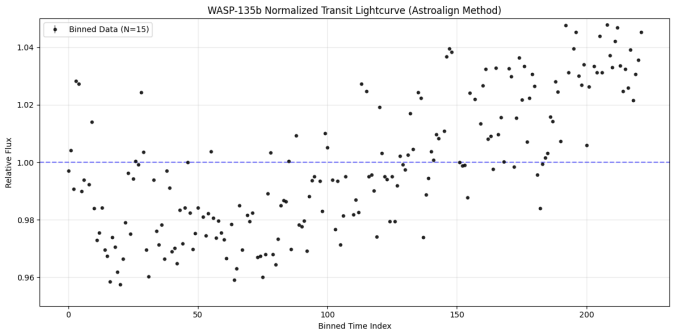



**Fig. 2:** Relative Flux vs Binned Time Index of WASP-135b, a bin size of 15 points was used. The blue line is 1.00 flux relative to its surroundings

The pipeline successfully recovered a transit, exhibiting a slight ingress, clear egress, and an out-of-transit baseline. However, the initial data remained highly scattered. To mitigate this noise while retaining sharp transit features, we applied a bin size of 15 points, meaning we averaged 15 consecutive flux values into a single data point. This was found to be most appropriate for mitigating noise while still retaining sharp transit features such as ingress and egress. Using the `batman` package, we made an initial guess (Table 2) and fitted a Maximum Likelihood Estimation model to the light-curve.

**Table 2:** Initial Guess Parametrization of Light-Curve

| Parameter | Initial Guess |
|---|---|
| $t_0$ | $t_{\text{lowest flux}}$ |
| $P$ | 1.401 |
| $R_p/R_s$ | 0.13 |
| $a/R_s$ | 8.0 |
| $i$ | 87° |
| $e$ | 0 |
| $\omega$ | 90 |
| $u_1$ | 0.3 |
| $u_2$ | 0.2 |

Once again, the fitted model somewhat successfully captures the transit, represented by the dip in flux corresponding to WASP-135b passing in front of WASP-135. However, the observational data continues upward in flux until the end, which is not accounted for in the current model. As a result, the fit deviates from the data outside the transit region. Nevertheless, we continue and apply MCMC analysis to our model and generate the corner plot to analyze our parameters.

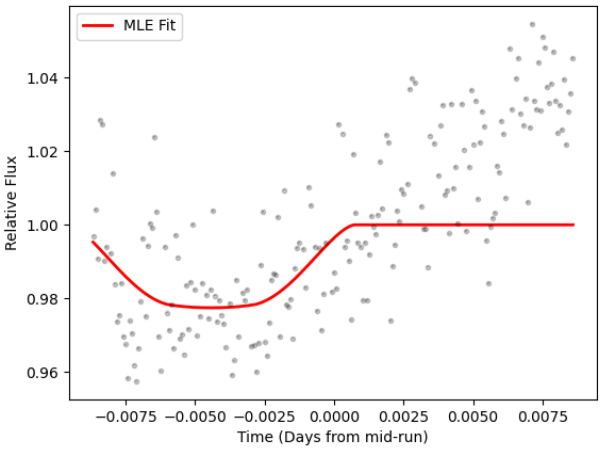



**Fig. 3:** Batman generated model using Table 2 values and then MLE fitted

**Table 3:** MCMC Priors for Each Model Parameter

| Parameter | Minimum | Maximum | Prior Type |
|---|---|---|---|
| $t_0$ | $t_{min}$ | $t_{max}$ | Uniform |
| $P$ | 1 | 5 | Log-uniform |
| $R_p$ | 0.05 | 2 | Uniform |
| $a$ | 1.0 | 20.0 | Uniform |
| $i$ | 70° | 90° | Uniform |
| $e$ | 0 | 0.3 | Uniform |
| $\omega$ | 0.0 | 360 | Uniform |
| $u_1$ | 0.0 | 1.0 | Gaussian |
| $u_2$ | 0.0 | 1.0 | Gaussian |

Prior types indicate how likely a value is to appear in the given region. Uniform assumes all values in a range are equally likely with outside values being rejected. Log-uniform is the same as uniform but in the log-space. Gaussian assumes value are most likely near a mean value with a smooth penalty for being further away.

Figure 4 presents the marginalized posterior probability distributions for the fitted transit parameters. This 'corner plot' visualizes the results of our MCMC analysis. The uppermost diagonal represents the distribution of values for each parameter, higher being more likely. If the distribution is bell shaped, it indicates that the parameter is well constrained by the data, with a clear central value and symmetrical uncertainties. In contrast, a heavily skewed distribution suggests that the parameter is either poorly constrained by the observations or is limited by the priors. The other plots represent the joint probability distribution between pairs of parameters. Circular contours mean the parameters are independent of each other while ellipses indicate there is some correlation between the pair. The highest probability values and their $\pm 1\sigma$ uncertainties are presented in Table 4.

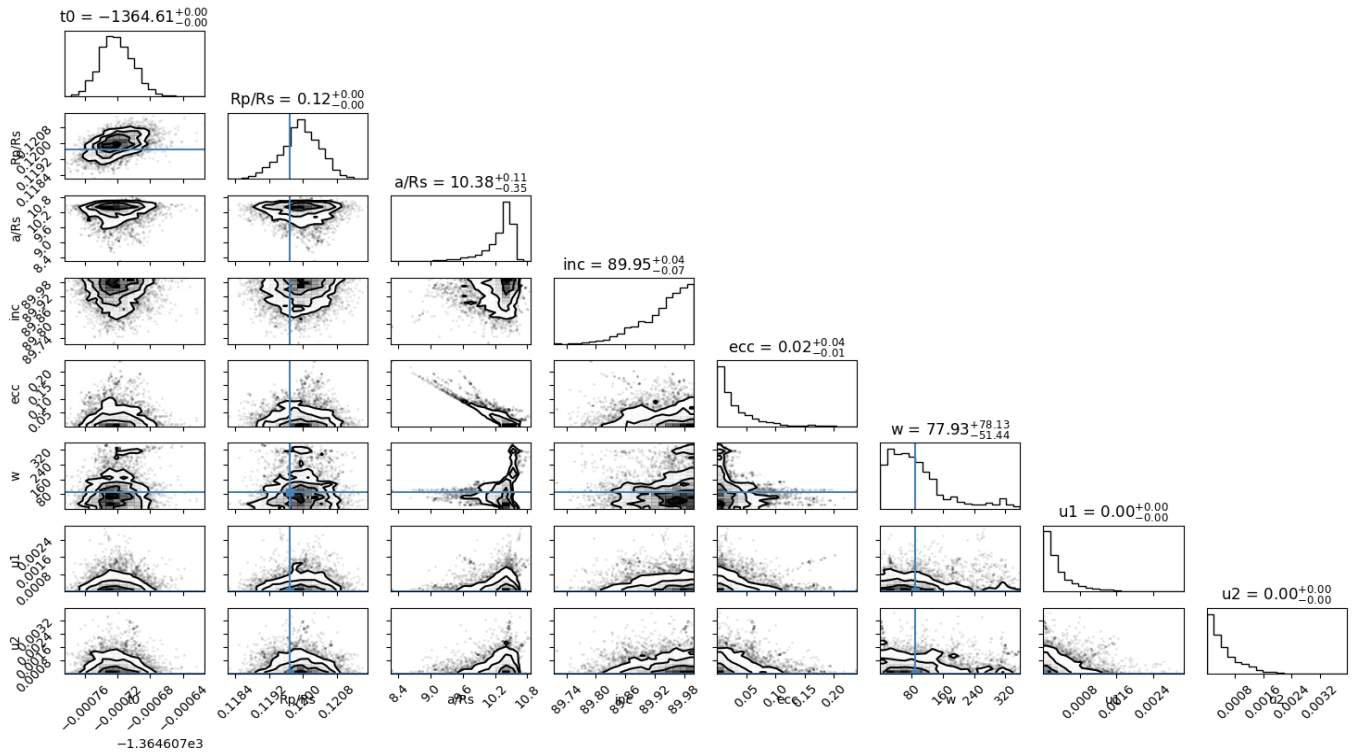



**Fig. 4:** Posterior distributions and correlations for the fitted WASP-135b parameters. The 1D histograms show the marginal distributions, while the 2D contour plots illustrate correlations between parameters. The blue lines represent the 16th, 50th and 84th percentiles. Plot generated using `corner.py` (Foreman-Mackey 2016).

**Table 4:** MCMC Analyzed Median Values of Each Model Parameter

| Parameter | Median | $-1\sigma$ | $+1\sigma$ |
|---|---|---|---|
| $t_0$ | -1364.60772 | -0.00002 | 0.00002 |
| $R_p/R_s$ | 0.11996 | -0.00045 | 0.00045 |
| $a/R_s$ | 10.37716 | -0.35222 | 0.11131 |
| $i$ | 89.94975° | -0.06570° | 0.03554° |
| $e$ | 0.01923 | -0.01449 | 0.04084 |
| $\omega$ | 77.92862 | -51.44138 | 78.13121 |
| $u_1$ | 0.00021 | -0.00016 | 0.00042 |
| $u_2$ | 0.00035 | -0.00025 | 0.00063 |

## 4. Discussion

### 4.1. Analysis of Corner Plot

Figure 4 reveals several physical realities of the WASP-135b system. The transit center ($t_0$) and radius ratio ($R_p/R_s$) present well defined, bell shaped distributions, proving the MCMC is very certain about these values. This is due to the fact that the depth and timing of the dip are the most prominent features in our light-curve.

On the other hand, all the other distributions are heavily skewed towards the boundaries. The orbital inclination ($i$) and eccentricity ($e$) pile up near 90° and 0 respectively. This aligns with expected orbital dynamics of a Hot Jupiter as tidal forces from the host star will have circularized its orbit over time, hence an eccentricity of zero. Furthermore, for us to see a transit of a planet at all, the system must be viewed almost perfectly edge-on (90°). The longitude of periastron ($\omega$) is neither bell shaped nor heavily leaning towards one side. Mathematically, this makes sense as the orbit is a near perfect circle ($e = 0$) so there is no closest point to the star. Thus, the MCMC couldn't find a single preferred value.

Lastly, the limb darkening coefficients ($u_1, u_2$) trend towards zero. The high scatter in our light-curve obscures the curvature of the transit, preventing the MCMC from distinguishing between background noise and the true limb darkening profile.

### 4.2. Comparison with Literature

Table 5 presents published values from the American Association of Variable Star Observers (AAVSO), an organization that collects and curates exoplanet modeling parameters submitted by the scientific community. Observation dates close to ours were chosen to maintain temporal consistency.

Comparing these values, only the radius ratio and second limb darkening coefficients fall within the uncertainties of the reference values. The agreement in the former is expected because it is derived directly from the transit depth. Even with our noisy data, there is a clear out-of-transit baseline and a clear bottom of the transit, for which the MCMC can accurately calculate the percentage of light blocked. The agreement in the second limb darkening coefficient, however, is most likely a mathematical artifact of the large uncertainties driven by the noisy data, rather than a strict physical constraint.

The significant difference in $T_0$ is likely not a failure of the fit, but a reflection of the differing frameworks between our observation and AAVSO's observation. The primary driver for the divergence in the remaining data is most likely the incomplete light-curve. Without a completely observed ingress, such as the one in figure 5, the MCMC sampler lacks a crucial constraint on when the transit event began. Hence, the model slides the transit center around to fit the observed egress, artificially adjusting

**Table 5:** Comparison of our data with AAVSO published parameters of WASP-135b

| Observer | Observation Date | $T_0$ | $R_p/R_s$ | $a/R_s$ | $u_1$ | $u_2$ |
|---|---|---|---|---|---|---|
| Our Data | 2024-07-14 | -1364.6077 | 0.1199 | 10.377 | 0.00021 | 0.0003 |
| RHEC | 2024-06-16 | 2460477.7509 | 0.1349 | 5.608 | 0.9631 | 0.0309 |
| NANF | 2024-06-23 | 2460484.7570 | 0.1754 | 5.240 | 0.995 | 0.0030 |
| BIRC | 2024-07-21 | 2460512.7818 | 0.1203 | 5.750 | 1.026 | -0.0210 |

**Notes.** [(a)] Values retrieved from the AAVSO Exoplanet Database (2026)

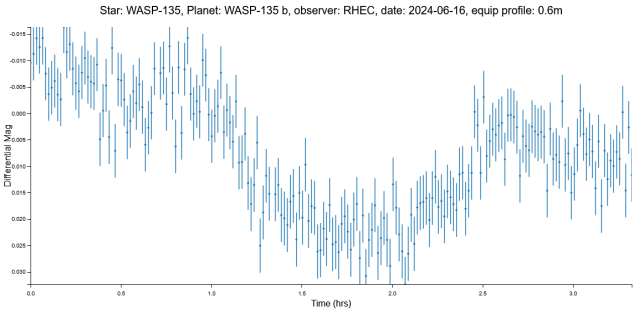



**Fig. 5:** AAVSO published light-curve. Image adapted from the AAVSO Exoplanet Database (2026)

the semi-major axis to match the exit slope. This effect is further amplified by the high level of scatter in the light curve. Features such as the sharp corners of the ingress/egress and the U-shape at the bottom of the transit are effectively washed out by the noise. Because these features are necessary to constrain our transit duration and limb darkening coefficients, the model often drives these parameters toward their prior boundaries giving the values observed.

The priors we chose were also uninformative, meaning the algorithm is purely chasing the noise in our specific light-curve. AAVSO light-curves utilize informative priors which means it already knows what WASP-135b is supposed to look like and uses that knowledge to guide the fit. This acts like a rubber band, pulling the MCMC back toward the true value even if the noise suggests otherwise.

### *4.3. Pipeline Analysis*

To verify that the parameter discrepancies were a result of data quality rather than pipeline architecture, we applied our MCMC code to a clean AAVSO light-curve (Figure 5). The informative priors included along with the model were used instead of our old uniform ones.

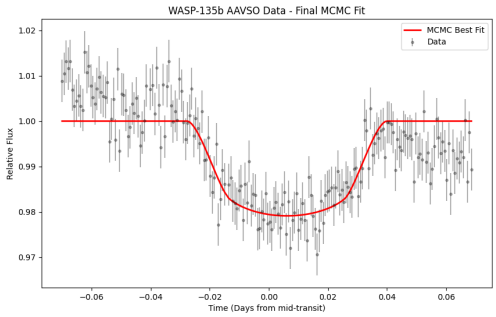



**Fig. 6:** Generated model of figure 5 light-curve using our MCMC code

Figure 6 shows the algorithm successfully generated a model with a distinct ingress and egress that closely tracks the relative flux. Figure 7 shows the resulting corner plot which displays well defined bell shaped distributions for the primary parameters, indicating that the walkers converged on a definitive solution. The newly derived values $R_p/R_s = 0.14, a/R_s = 5.53, u_1 = 0.54, u_2 = 0.10$ strongly agree with the AAVSO values in table 5, confirming the computational integrity of our modeling pipeline.

## 5. Conclusions

We present a comprehensive, Python based pipeline designed to process raw telescope images and model exoplanet transits using Maximum Likelihood Estimation and Markov Chain Monte Carlo methods. We applied this pipeline to a dataset of the Hot Jupiter WASP-135b. While the fundamental radius ratio was successfully recovered and aligned with literature values, high photometric scatter and incomplete transit coverage, specifically the lack of an ingress, severely limited the constraint of geometric parameters such as orbital duration and limb darkening coefficients. To validate the computational accuracy of our code, we subsequently analyzed a high accuracy AAVSO dataset, which successfully yielded parameter distributions correlated with published reference values. Ultimately, these results demonstrate that while robust analytical algorithms are essential for planetary characterization, the precision of derived physical parameters remains fundamentally constrained by the baseline quality and completeness of the observational data.

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## Data Availability

The raw photometric data for WASP-135b analyzed in this study were obtained from Professor Lea Hirsch. The comparison dataset and reference model parameters were retrieved from the AAVSO Exoplanet Database (AAVSO 2026). All processed light-curves and MCMC Python scripts are available upon request.

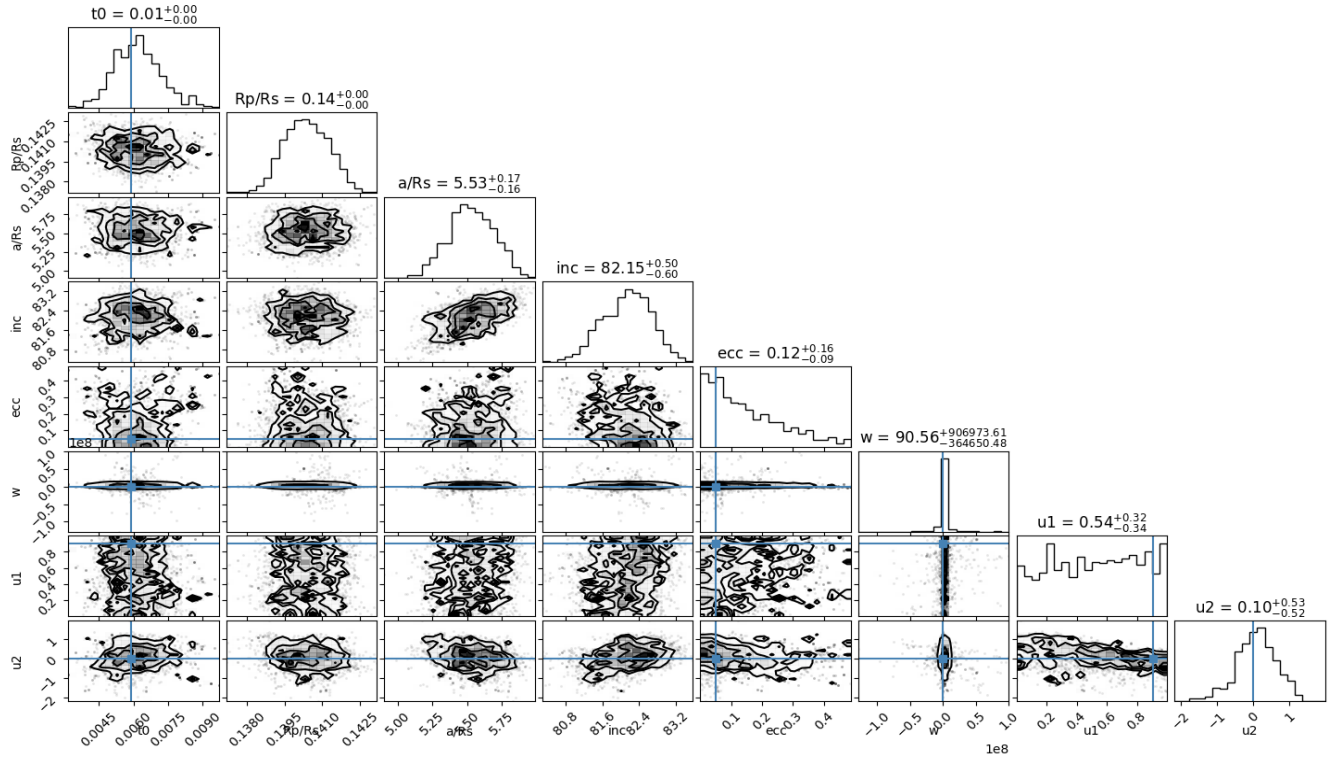



**Fig. 7:** Corner Plot of figure 5 light-curve using our MCMC code. Plot generated using `corner.py` (Foreman-Mackey 2016).

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